Smart Waste Collection and Recycling Management System

Pseudo code

```
START

Create an empty list called waste_records

REPEAT

  Display Menu

  1. Add Record
  2. Display Records
  3. Search Record
  4. Total Waste Analysis
  5. Save Records
  6. Load Records
  7. Exit

  Read user choice

  IF choice = 1 THEN

    Read House Number
    Read Waste Type
    Read Quantity

    Create tuple:
    record ← (House Number, Waste Type, Quantity)

    Add record to waste_records
```

```
    Display "Record Added Successfully"

ELSE IF choice = 2 THEN

    IF waste_records is empty THEN
        Display "No Records Available"
    ELSE
        FOR each record in waste_records DO
            Display House Number
            Display Waste Type
            Display Quantity
        END FOR
    END IF

ELSE IF choice = 3 THEN

    Read House Number to search

    FOR each record in waste_records DO

        IF House Number matches THEN

            Display Record Details

            Stop Search

        END IF

    END FOR
```

```
    If record not found
        Display "Record Not Found"

ELSE IF choice = 4 THEN

    total ← 0

    FOR each record in waste_records DO

        total ← total + Quantity

    END FOR

    Display Total Waste Collected

ELSE IF choice = 5 THEN

    Open file "waste_data.txt" in write mode

    FOR each record in waste_records DO

        Write House Number,
              Waste Type,
              Quantity into file

    END FOR

    Close file
```

```
        Display "Records Saved Successfully"

    ELSE IF choice = 6 THEN

        Open file "waste_data.txt" in read mode

        Read each line from file

        Convert data into tuple

        Store tuple in waste_records

        Close file

        Display "Records Loaded Successfully"

    ELSE IF choice = 7 THEN

        Display "Thank You"

        EXIT LOOP

    ELSE

        Display "Invalid Choice"

    END IF

UNTIL choice = 7
```

STOP

Module-wise Pseudocode

Add record

Input House Number

Input Waste Type

Input Quantity

Create Tuple

Store Tuple in waste_records

Search record

Input House Number

Compare with each record

If found display details

Else display Record Not Found

Save record

Open file

Write all tuples into file

Close file

Waste analysis

Initialize total = 0

Add quantity from each tuple

Display total waste

Sourse code

```
import customtkinter as ctk

from tkinter import messagebox


ctk.set_appearance_mode("light")

ctk.set_default_color_theme("blue")


# Store records as tuples

waste_records = []


# ---------------- FUNCTIONS ---------------- #


def add_record():

    house = house_entry.get().strip()

    waste_type = waste_entry.get().strip().title()

    quantity = quantity_entry.get().strip()


    if house == "" or waste_type == "" or quantity == "":

        messagebox.showerror("Error", "Please fill all fields")

        return


    try:

        quantity = float(quantity)

    except ValueError:

        messagebox.showerror("Error", "Quantity must be a number")

        return


    record = (house, waste_type, quantity)

    waste_records.append(record)
```

```
    messagebox.showinfo("Success", "Record Added Successfully")

    house_entry.delete(0, "end")
    waste_entry.delete(0, "end")
    quantity_entry.delete(0, "end")

    display_records()


def display_records():

    output_box.delete("1.0", "end")

    output_box.insert(
        "end",
        "House No\tWaste Type\tQuantity(kg)\n"
    )
    output_box.insert(
        "end",
        "---------------------------------------------\n"
    )

    for record in waste_records:
        output_box.insert(
            "end",
            f"{record[0]}\t\t{record[1]}\t\t{record[2]}\n"
        )
```

```
def search_record():

    house = house_entry.get().strip()

    for record in waste_records:

        if record[0] == house:

            messagebox.showinfo(
                "Record Found",
                f"House Number : {record[0]}\n"
                f"Waste Type : {record[1]}\n"
                f"Quantity : {record[2]} kg"
            )
            return

    messagebox.showerror("Not Found", "Record Not Found")



def total_waste():

    total = 0

    for record in waste_records:
        total += record[2]

    messagebox.showinfo(
        "Waste Analysis",
        f"Total Waste Collected = {total} kg"
    )
```

```
def save_records():

    with open("waste_data.txt", "w") as file:

        for record in waste_records:

            file.write(
                f"{record[0]},{record[1]},{record[2]}\n"
            )

    messagebox.showinfo(
        "Saved",
        "Records Saved Successfully"
    )


def load_records():

    waste_records.clear()

    try:

        with open("waste_data.txt", "r") as file:

            for line in file:

                house, waste_type, quantity = line.strip().split(",")
```

```
            record = (
                house,
                waste_type,
                float(quantity)
            )

            waste_records.append(record)

        display_records()

        messagebox.showinfo(
            "Loaded",
            "Records Loaded Successfully"
        )

    except FileNotFoundError:

        messagebox.showerror(
            "Error",
            "No Saved File Found"
        )

# ---------------- GUI ---------------- #

app = ctk.CTk()
app.geometry("850x700")
app.title("Smart Waste Collection System")

title = ctk.CTkLabel(
    app,
```

```
    text="Smart Waste Collection and Recycling Management System",
    font=("Arial", 24, "bold")
)
title.pack(pady=20)

# House Number Label
house_label = ctk.CTkLabel(
    app,
    text="House Number",
    font=("Arial", 16)
)
house_label.pack()

house_entry = ctk.CTkEntry(
    app,
    width=300
)
house_entry.pack(pady=10)

# Waste Type Label
waste_label = ctk.CTkLabel(
    app,
    text="Waste Type",
    font=("Arial", 16)
)
waste_label.pack()

waste_entry = ctk.CTkEntry(
    app,
    width=300
```

```
)
waste_entry.pack(pady=10)

# Quantity Label
quantity_label = ctk.CTkLabel(
   app,
   text="Quantity (kg)",
   font=("Arial", 16)
)
quantity_label.pack()

quantity_entry = ctk.CTkEntry(
   app,
   width=300
)
quantity_entry.pack(pady=10)

# Buttons
ctk.CTkButton(
   app,
   text="Add Record",
   command=add_record,
   width=200
).pack(pady=5)

ctk.CTkButton(
   app,
   text="Display Records",
   command=display_records,
   width=200
```

```
).pack(pady=5)

ctk.CTkButton(
    app,
    text="Search Record",
    command=search_record,
    width=200
).pack(pady=5)

ctk.CTkButton(
    app,
    text="Total Waste Analysis",
    command=total_waste,
    width=200
).pack(pady=5)

ctk.CTkButton(
    app,
    text="Save Records",
    command=save_records,
    width=200
).pack(pady=5)

ctk.CTkButton(
    app,
    text="Load Records",
    command=load_records,
    width=200
).pack(pady=5)
```

```
# Output Area

output_box = ctk.CTkTextbox(

    app,

    width=700,

    height=250

)

output_box.pack(pady=20)


app.mainloop()
```

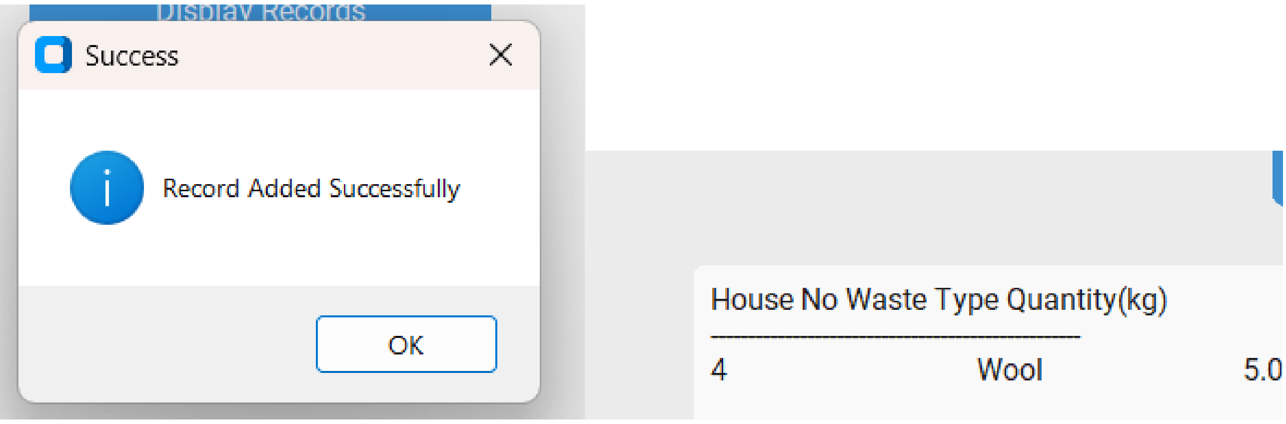
Output screenshot

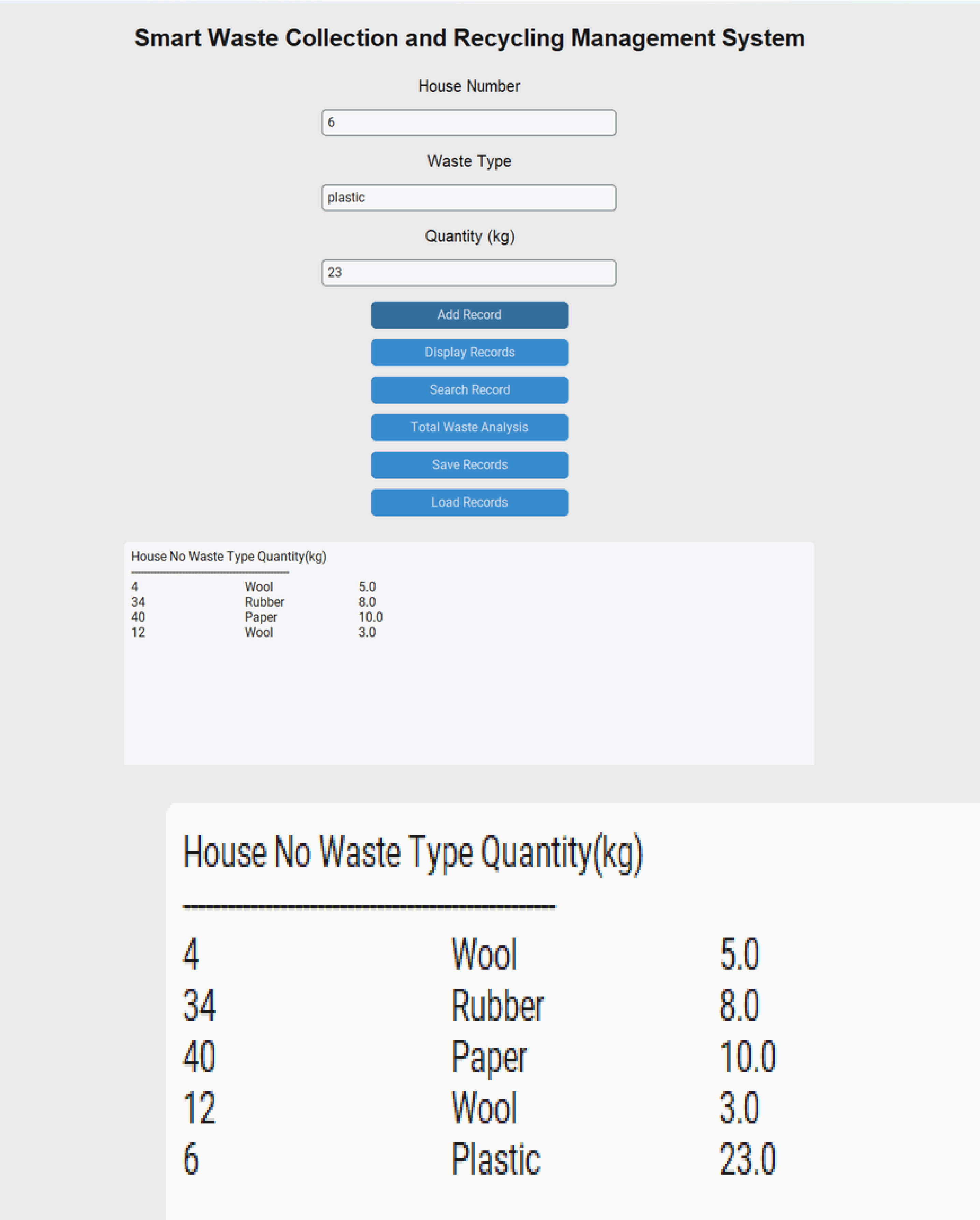
python assignment project.py - C:/Users/smohi/AppData/Local/Programs/Python/Python312/python assignment project.py (3.12.7)

File Edit Format Run Options Window Help

```python
from tkinter import messagebox

ctk.set_appearance_mode("light")
ctk.set_default_color_theme("blue")

# Store records as tuples
waste_records = []

# ---------------- FUNCTIONS ---------------- #

def add_record():
    house = house_entry.get().strip()
    waste_type = waste_entry.get().strip().title()
    quantity = quantity_entry.get().strip()

    if house == "" or waste_type == "" or quantity == "":
        messagebox.showerror("Error", "Please fill all fields")
        return

    try:
        quantity = float(quantity)
    except ValueError:
        messagebox.showerror("Error", "Quantity must be a number")
        return

    record = (house, waste_type, quantity)
    waste_records.append(record)

    messagebox.showinfo("Success", "Record Added Successfully")

    house_entry.delete(0, "end")
    waste_entry.delete(0, "end")
    quantity_entry.delete(0, "end")

    display_records()


def display_records():

    output_box.delete("1.0", "end")

    output_box.insert(
        "end",
        "House No\tWaste Type\tQuantity(kg)\n"
    )
    output_box.insert(
        "end",
        "--------------------------------------------------\n"
    )
```

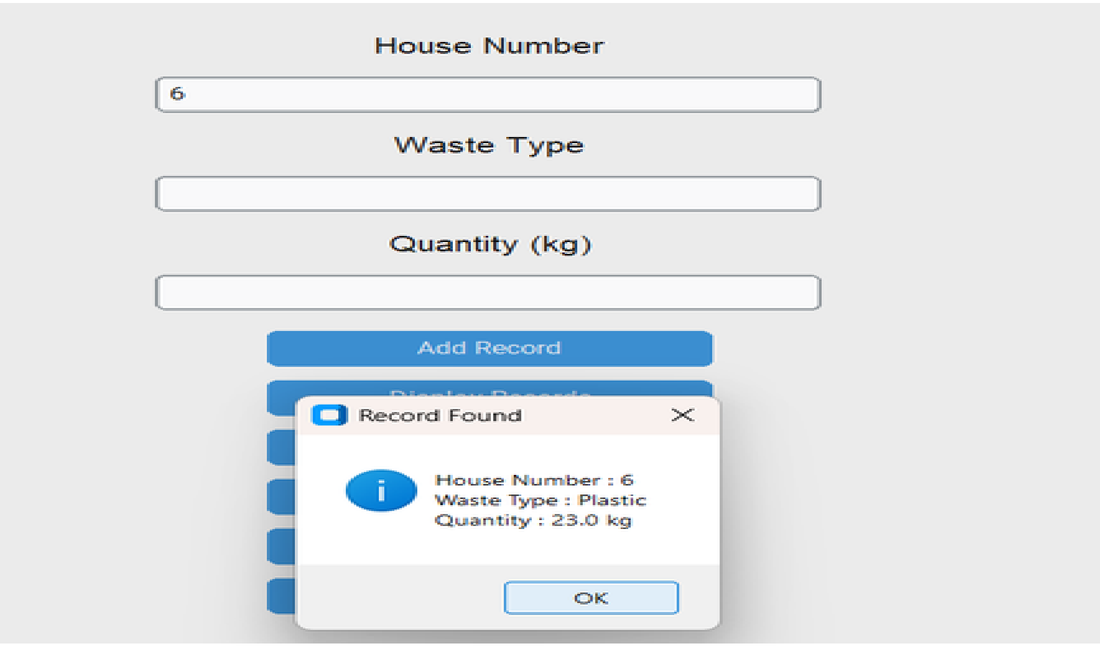
Record adding

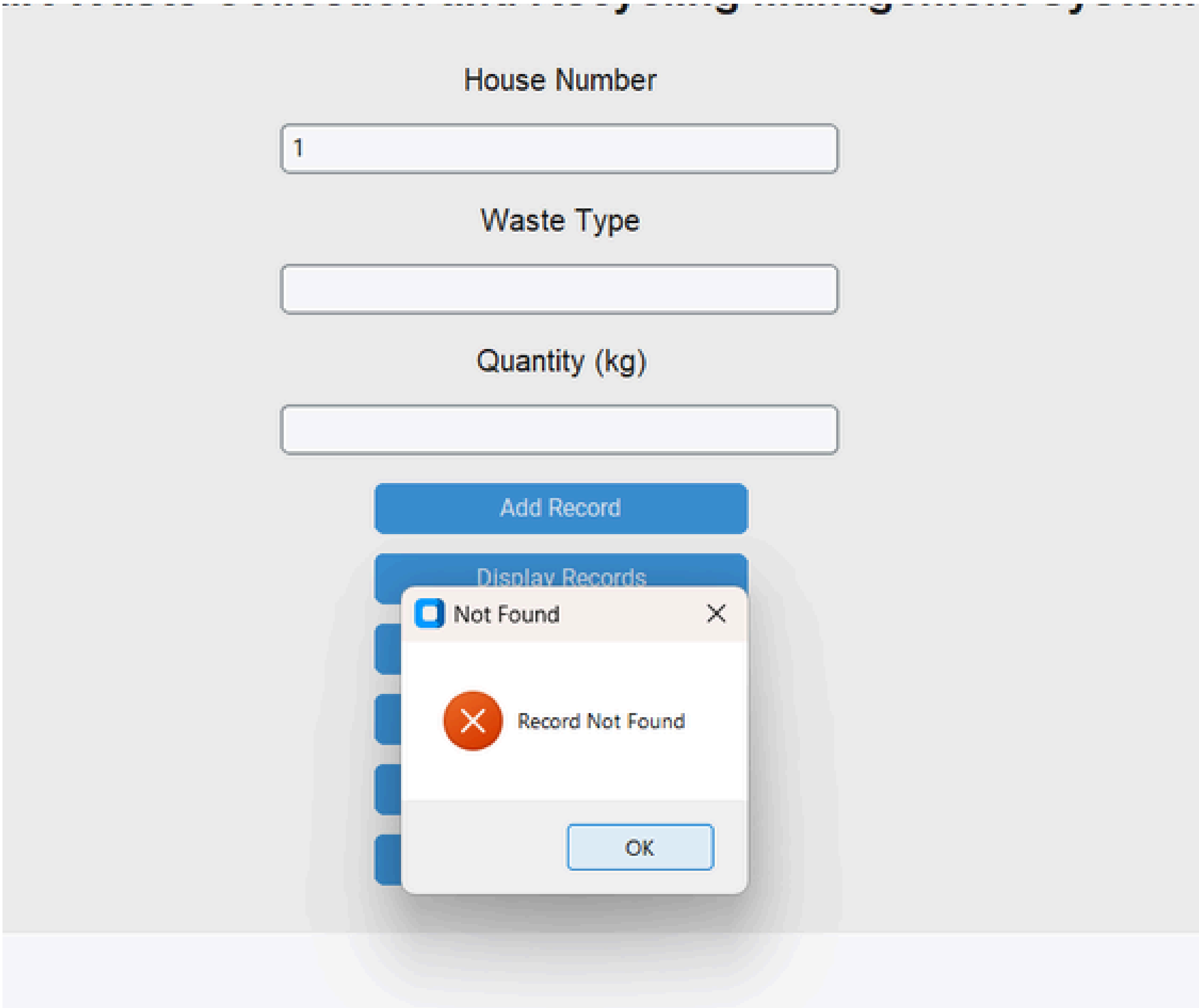




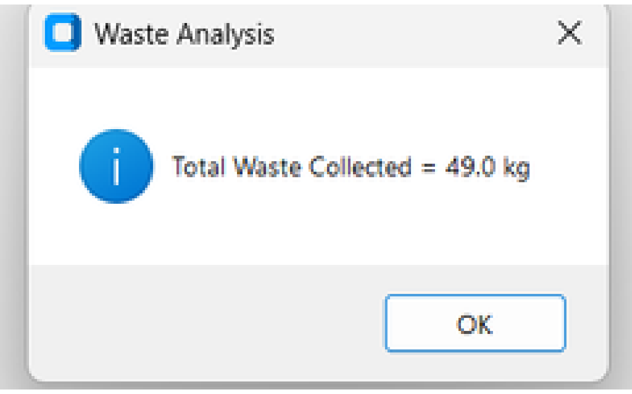
Display record

Search record



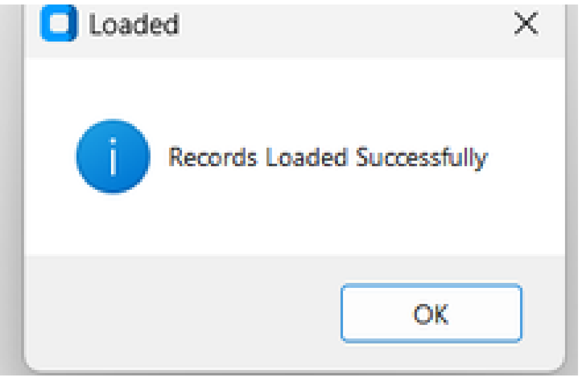


Total waste analysis in a single day



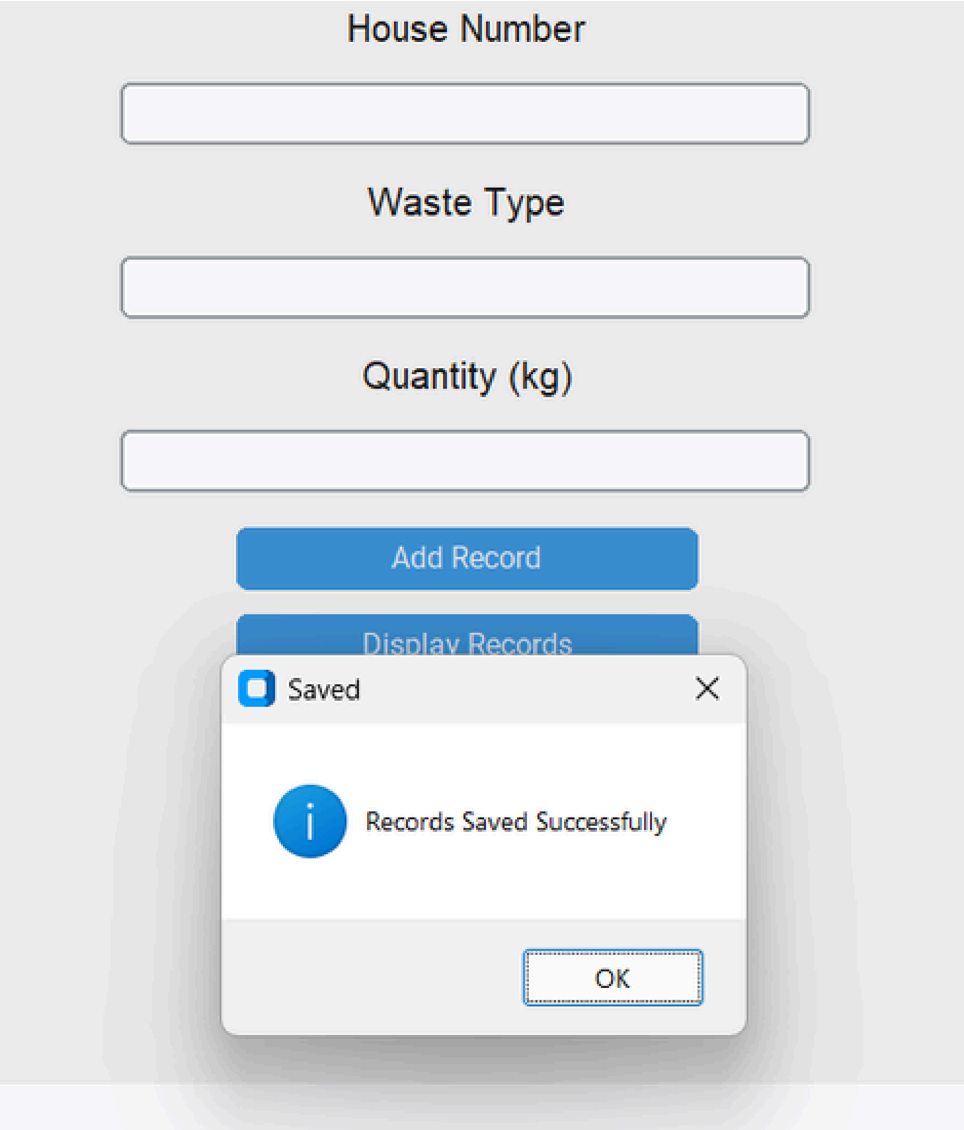
Load record

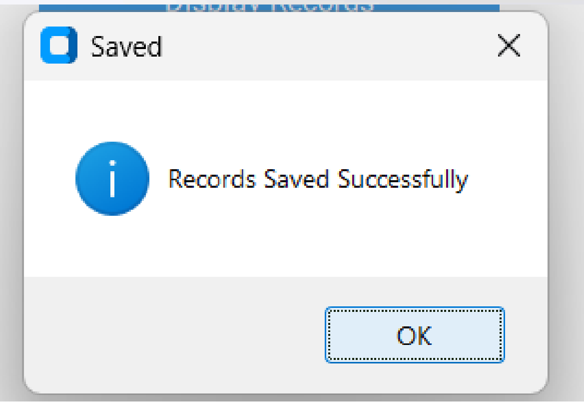

Smart Waste Collection and Recycling Management System
House Number
Waste Type
Quantity (kg)
Add Record
Display Records
Search Record
Total Waste Analysis
Save Records
Load Records


Loaded
Records Loaded Successfully
OK

```
House No Waste Type Quantity(kg)
------------------------------------------------
4          Wool       5.0
34         Rubber     8.0
40         Paper      10.0
12         Wool       3.0
6          Plastic    23.0
```

Save record


House Number
Waste Type
Quantity (kg)
Add Record
Display Records
Saved
Records Saved Successfully
OK


Saved
Records Saved Successfully
OK